

# Applied Statistical Analysis I

## Regression Diagnostics

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Elena Karagianni, PhD Candidate

karagiae@tcd.ie

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Department of Political Science, Trinity College Dublin

## OLS assumptions

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What are the assumptions of linear regression?

Assumptions about the model specification  $Y_i = \alpha + \beta X_i + \epsilon_i$

- **Linearity** in parameters. Our model can be written as:

$$Y = \alpha + \beta_1 X_1 + \beta_2 X_2 + \dots + \beta_k X_k + \epsilon$$

- **X values are measured without error**, the variability is due to the stochastic component (i.e., in  $\epsilon$ ).
- No causal variables are left out while non-causal variables included.

# OLS assumptions

## Assumptions about the error ( $\epsilon_i$ )

- The **zero conditional mean assumption** requires that:

$$E(\epsilon|x_1, x_2, \dots, x_k) = 0$$

The error term is unrelated to the regressors  $\rightarrow$  violated by omitted variables, reverse causality, or measurement error in  $X$ .

- Error variance is constant given the data (**homoscedasticity assumption**).

$$\text{Var}(\epsilon|x_1, x_2, \dots, x_k) = \sigma_\epsilon^2$$

- Errors are normally distributed,  $\epsilon_i \sim N(0, \sigma^2) \rightarrow$  needed for inference
- Errors for any two observations are uncorrelated (**independence assumption**).  
Hence,  $\text{Cov}(\epsilon_i, \epsilon_j|x_i, x_j) = 0, i \neq j$

# Mathematical preconditions for OLS

- $X$  must vary
- Number of observations must be larger than the number of predictors
- In multiple regression: no perfect multicollinearity

# Violations

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# Multicollinearity

- Multicollinearity refers to high correlation between two or more independent variables. It is not limited to pairwise correlation.
- But multicollinearity can lead to instability in the regression estimates - the same consequence that follows from having a small sample.
- When the covariates are strongly correlated, the data contain little information about the impact of, say,  $x_1$ , on  $Y$  holding  $x_2, \dots, x_k$  constant, because there is little variation in  $x_1$ , when  $x_2, \dots, x_k$  are fixed.
- Strong, but less than perfect, multicollinearity substantially increases the sampling variances of the OLS coefficients. Consequently, this also increases CIs and SEs of the coefficients. Thus, it also affects hypothesis testing.
- The good news is: even strong multicollinearity does not bias our coefficients!



## Multicollinearity diagnosis

- A **correlation matrix** gives you a first idea. Pairwise correlations cannot reveal highly collinear **linear combinations**, though.
- A good strategy to detect it is to calculate the **Variance-Inflation Factor (VIF)** for each covariate, which is defined as

$$\text{VIF}_j = \frac{1}{(1 - R_j^2)},$$

where  $R_j^2$  is again the  $R^2$  measure from a regression of  $x_j$  on the remaining covariates.

- A problem with VIF is that we need to determine an arbitrary threshold above which we consider multicollinearity to be high.
- Sometimes, 10 is chosen as this threshold value.

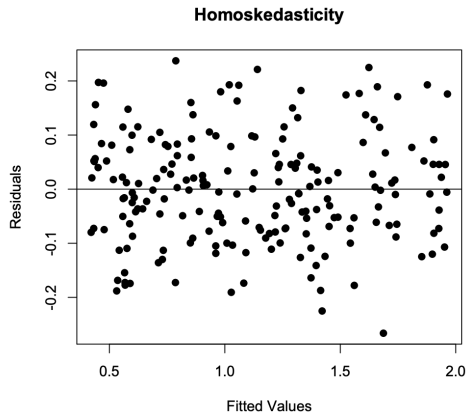
## Multicollinearity Remedies

- There is no good way to reduce variances of unbiased estimators other than to collect more data (increase the sample size).
- Possibly simplify model, i.e., drop variables to reduce collinearity. Unfortunately, dropping relevant variables leads to omitted variable bias.
- Possibly combine variables into one index. This way we would no longer be trying to estimate the partial effect of each separate category.
- If we believe certain variables belong in the model (e.g., to infer causality), then there is not much we can do about it. Our theories should tell us what to include, but we may be able to use different operationalizations.
- If we expect to see an effect, but CIs appear too wide, checking VIF is a good diagnostic tool.

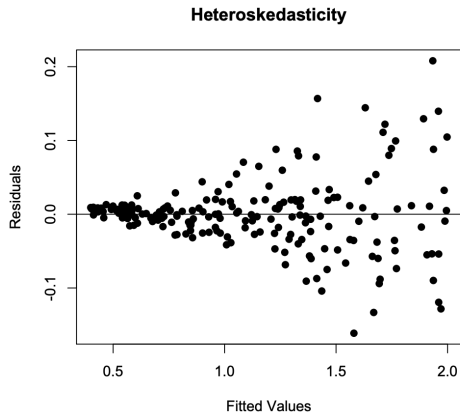
# Heteroscedasticity

- The assumption of **homoscedasticity** is necessary to justify the t-tests and CIs for OLS estimation.
- But: it is common for the variance of errors to increase or decrease with the level of the independent variable.
- Such a pattern is called **heteroscedasticity**. OLS standard errors are no longer valid for constructing CIs or hypothesis testing (they are unreliable).
- Heteroscedasticity can be detected in a plot of residuals against fitted values.

# Heteroskedasticity



What we want



This is a problem

# Heteroskedasticity

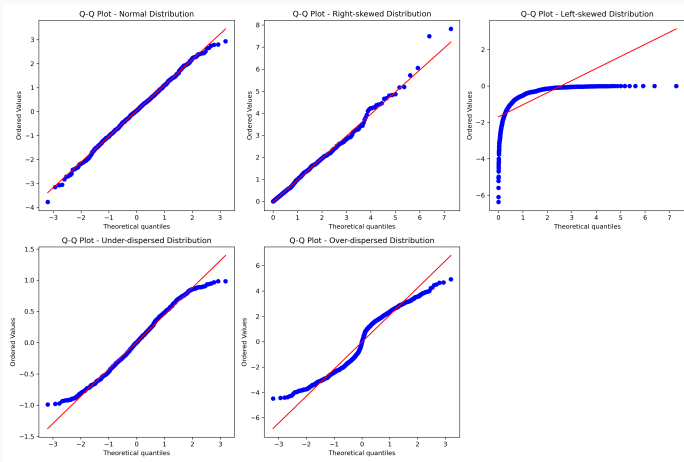
- The error variance might vary because of *model specification errors* (i.e., the model is not correctly specified).
  - Transform your independent variables and potentially the dependent variable. Plot residuals against the respective variables to find candidates for this.
  - Or include interaction effects.
  - Use robust standard errors.
- Heteroskedasticity can arise as a result of the presence of **outliers**.

# Normal distribution of $\epsilon_i$

Validate through:

- Histogram for  $\epsilon_i$
- QQ plot

If this assumption is violated  $\rightarrow$  t-tests and CIs may be unreliable in small sample. You can fix it by gathering more data (increasing the sample size).



# Parametric linearity

Validate through:

1. Scatter plot (for simple regression)
2. Residual vs. fitted plot

If violated → slope coefficients are biased and predictions unreliable.

Remedies:

- Transform variables (log, square root, polynomial terms)
- Include interaction terms

## Non-linear effects: An example

Model a curvilinear (=curved lines) relationship between an independent variable and the dependent variable.

Include X **and** the square of X:

$$\hat{Y}_i = \alpha + \beta_1 X_i + \beta_2 X_i^2 + \epsilon_i$$



# Non-linear effects: An example

“U-shaped” relationship between democracy and environment protection?

```
1 # Generate quadratic term
2 qog_data$sqr_vdem_polyarchy <- qog_data$vdem_polyarchy^2
3
4 # Run ols regression with quadratic term
5 q_m1 <- lm(epi_epi ~ income + vdem_polyarchy
6           + sqr_vdem_polyarchy,
7           data = qog_data)
8
9 # Print results
10 summary(q_m1)
```

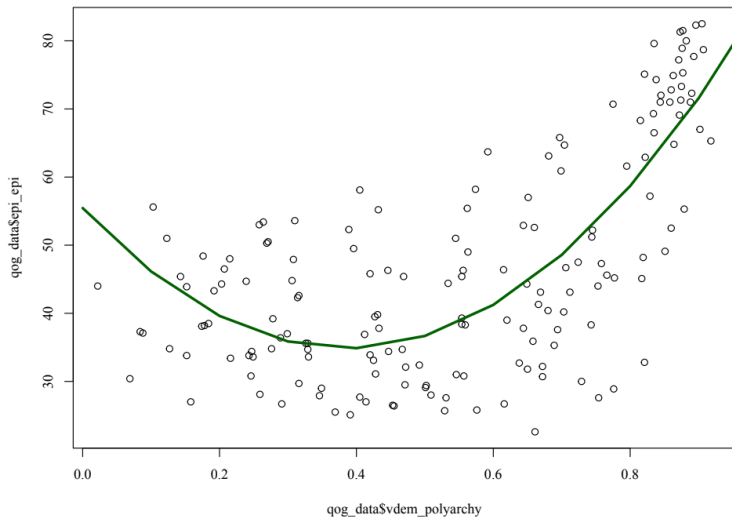
|                    | Estimate | Std. Error | t value | Pr(> t ) |     |
|--------------------|----------|------------|---------|----------|-----|
| (Intercept)        | 39.4244  | 4.2944     | 9.180   | 2.82e-16 | *** |
| income             | 3.0094   | 0.4576     | 6.576   | 7.19e-10 | *** |
| vdem_polyarchy     | -44.3531 | 17.7037    | -2.505  | 0.0133   | *   |
| sqr_vdem_polyarchy | 74.1559  | 17.0553    | 4.348   | 2.50e-05 | *** |

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Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 9.133 on 153 degrees of freedom  
(37 observations deleted due to missingness)  
Multiple R-squared: 0.6819, Adjusted R-squared: 0.6757  
F-statistic: 109.3 on 3 and 153 DF, p-value: < 2.2e-16

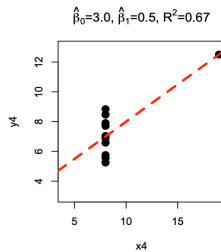
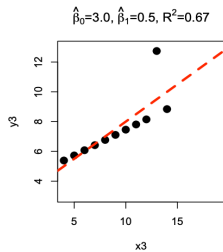
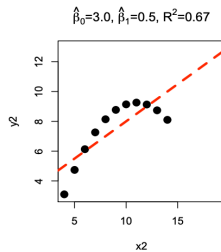
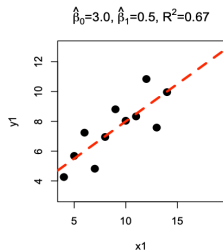
## Non-linear effects: An example



# Outliers

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# Influential observations

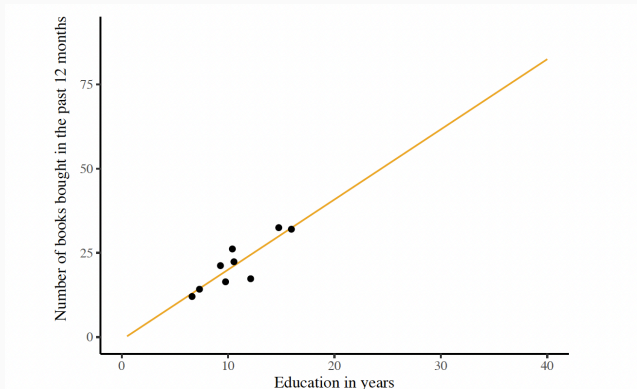


# Influential observations

- The above examples show that OLS estimation can be very sensitive to singletons.
- Therefore, it is important:
  - to identify observations that may be dominating the estimates, and
  - to assess how results change when such dominating data points are dropped.
- Some terminology:
  - **Outlier**: An observation with a **large residual**; it may indicate a sample peculiarity, a data entry error, or some other problems.
  - **Leverage**: An observation with an extreme value on an independent variable is said to have **high leverage**. Leverage is a measure of how far a covariate deviates from its mean. These leverage points may have an effect on the estimation of regression coefficients.
  - **Influence**: An observation is said to be influential if removing the observation **substantially changes** the estimate of coefficients.

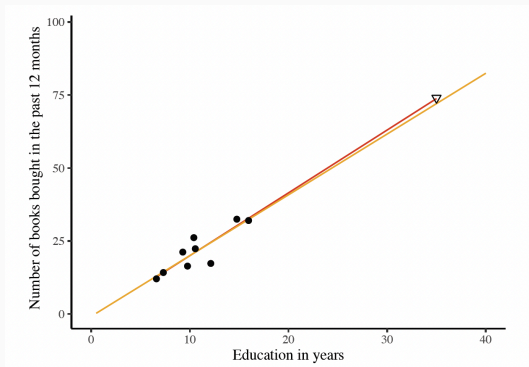
# Discrepancy, Leverage, and Influence

Not all outliers are concerning, because leverage  $\neq$  influence, and discrepancy  $\neq$  influence  $\rightarrow$  Influence is a joint function of leverage and discrepancy.



# Leverage

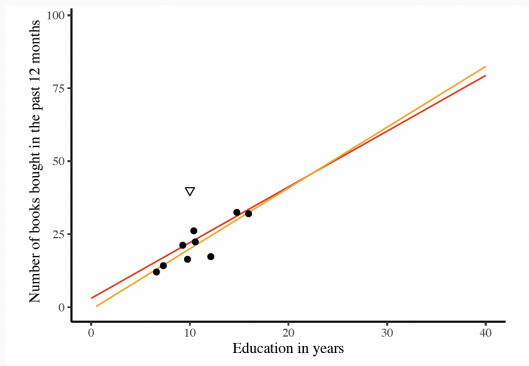
Observation is unusual in its value on X, has high leverage, but low discrepancy → low influence.



→ Hat values ( $h_i$ ), distance of each observation from the data center.

# Discrepancy

Observation is unusual in its value on Y, given its value on X, has high discrepancy, but low leverage. → low influence.

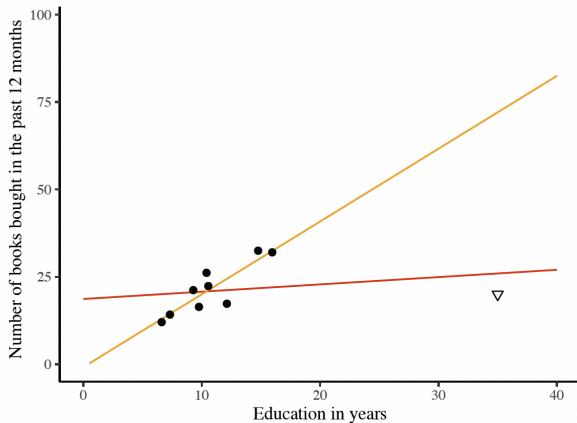


→ Standardized ( $\hat{\epsilon}_i'$ ) and studentized residuals ( $\hat{\epsilon}_i^*$ ), because  $\epsilon_i$  is scale-dependent and high leverage leads to low  $\epsilon_i$



# Influence

Observation has high leverage and discrepancy, an unusual value of X and Y  $\rightarrow$  high influence.



Validate through:

1. Cook's distance
2. Difference in betas (DFBeta), difference in coefficients when observation  $i$  is included and not included.
3. Leverage vs. residual plot

Remedies:

- Check for coding errors
- Think carefully about omitted variables

## Cook's Distance

- The Cook's distance is a **measure of influence**, which aggregates outlier and leverage properties.
- For each observation  $i$  Cook's distance is defined as:

$$D_i = \frac{\sum_j (\hat{y}_j - \hat{y}_{j(i)})^2}{(k+1) \hat{se}^2}$$

where  $\hat{y}_j$  is the predicted value for observation  $j$  with all data,  $\hat{y}_{j(i)}$  is the predicted value for  $j$  when observation  $i$  is dropped,  $k$  is the number of covariates, and  $\hat{se}^2$  is the estimated error variance.

- The measure of Cook's distance is based on the **difference in predicted values** with and without particular observations.
- Cook's distance is **model specific**, i.e., you cannot compare values from different models.

- The rule of thumb is to be suspicious of observations with  $D_i > 1$ .
- Bollen and Jackman (1990) suggest to determine the threshold as  $4/n$  where  $n$  is the sample size.